

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 24: Measures of Central Tendency (Mean, Median, Quartiles and Mode)

Every formula and result of the chapter on one sheet

1 Mean of raw data

$$\bar{x} = \frac{\sum x}{n}$$

2 Mean: direct method

$$\bar{x} = \frac{\sum fx}{\sum f}$$

3 Mean: short-cut method

$$\bar{x} = A + \frac{\sum fd}{\sum f}, \text{ where } d = x - A$$

4 Mean: step-deviation method

$$\bar{x} = A + \frac{\sum ft}{\sum f} \times i, \text{ where } t = \frac{x - A}{i}$$

5 Sum of the observations

$$n \times \bar{x}$$

6 Median, n oddThe $\left(\frac{n+1}{2}\right)$ th term7 Median, n evenMean of the $\left(\frac{n}{2}\right)$ th and $\left(\frac{n}{2} + 1\right)$ th terms8 Lower quartile Q_1 n odd: $\left(\frac{n+1}{4}\right)$ th term; n even: $\left(\frac{n}{4}\right)$ th term9 Upper quartile Q_3 n odd: $\left(\frac{3(n+1)}{4}\right)$ th term; n even: $\left(\frac{3n}{4}\right)$ th term

10 Inter-quartile range

$$Q_3 - Q_1$$

11 Semi-inter-quartile range

$$\frac{Q_3 - Q_1}{2}$$

12 Mode

The value with the highest frequency

13 From an ogive

Median at $\frac{N}{2}$, Q_1 at $\frac{N}{4}$, Q_3 at $\frac{3N}{4}$